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CKD and the Risk of Incident Cancer

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ABSTRACT

Previous studies report a higher risk of cancer in patients with ESRD, but the impact of less severe CKD on risk of cancer is uncertain. Our objective was to evaluate the association between level of kidney function and subsequent cancer risk. We performed a retrospective cohort study of 1,190,538 adults who were receiving care within a health care delivery system, had a measurement of kidney function obtained between 2000 and 2008, and had no prior cancer. We examined the association between level of eGFR and the risk of incident cancer; the primary outcome was renal cancer, and secondary outcomes were any cancer and specific cancers (urothelial, prostate, breast, lung, and colorectal). During 6,000,420 person-years of follow-up, we identified 76,809 incident cancers in 72,875 subjects. After adjustment for time-updated confounders, lower eGFR (in milliliters per minute per 1.73 m^2) was associated with an increased risk of renal cancer (adjusted hazard ratio [HR], 1.39; 95% confidence interval [95% CI], 1.22 to 1.58 for eGFR=45–59; HR, 1.81; 95% CI, 1.51 to 2.17 for eGFR=30–44; HR, 2.28; 95% CI, 1.78 to 2.92 for eGFR<30). We also observed an increased risk of urothelial cancer at eGFR<30 but no significant associations between eGFR and prostate, breast, lung, colorectal, or any cancer overall. In conclusion, reduced eGFR is associated with an independently higher risk of renal and urothelial cancer but not other cancer types.

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CKD and cancer are both major and growing public health problems nationally and internationally.[1,2] The incidence of CKD continues to rise, with an estimated 11.5% of the United States population having reduced eGFR and/or proteinuria and approximately 13.5 million Americans with stage 3 or higher CKD.[1,3] Concurrently, >1.5 million new cancer diagnoses and approximately 577,000 cancer deaths are projected nationally in 2012.[2] Although cancer incidence has decreased slightly since 2000, it remains one of the leading causes of morbidity and mortality.

The link between CKD and the risk of developing cancer has not been well delineated. Although multiple studies have observed higher risks of cancer in persons with ESRD requiring dialysis or renal transplantation,[4–11] whether less severe kidney disease is associated with cancer remains poorly understood.[12–15] Previous studies are limited by modest sample sizes, restricted diversity, and their ability to adequately control for confounders. Determining whether there is a robust association between the presence and severity of CKD with subsequent cancer risk and specifically examining if level of kidney function is differentially associated with specific cancer type could have important public health implications for screening and early detection of cancer in patients with CKD.

To address this knowledge gap, we evaluated the association between level of kidney function and the risk of incident cancer and cancer type within a large,

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diverse, community-based population linked to a regional cancer registry. We hypothesized that there would be an independent increased risk of kidney cancer with lower GFR levels but no significant relationship between renal function and other types of malignancies.

RESULTS

We identified 1,190,538 adults ages 40 years and older with known kidney function and no history of cancer, dialysis, or renal transplantation. Median follow-up for the cohort was 5.3 years (interquartile range=2.6–7.6 years), with a total of 6,000,420 person-years of follow-up. During follow-up, the median number of outpatient serum creatinine measurements per subject was four (interquartile range=2–9).

At cohort entry, patients with lower GFR were more likely to be older, be persons of color, have lower socioeconomic status, have a higher burden of comorbidity, and be current or former smokers (Table 1). The frequency of documented urine dipstick proteinuria was substantially higher at lower entry GFR levels along with the receipt of angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, aldosterone receptor antagonists, α-blockers, β-blockers, calcium channel blocker, statins, other lipid-lowering agents, and 5-α reductase inhibitors. Approximately 34% of the cohort had an abnormal dipstick hematuria evaluation and 30% obtained abdominal imaging during follow-up. However, the median number of ambulatory health care visits during the year before cohort entry was similar across all GFR categories (Table 1).

During follow-up, 81,545 subjects died, 4389 subjects initiated chronic dialysis, 1436 subjects received a renal transplant, and 296,734 subjects disenrolled from the health plan. During follow-up, in total, 76,809 incident cancers were documented among 72,875 subjects (38,744 men and 34,131 women). The age–sex-adjusted rates of incident renal cancer increased with lower GFR level, ranging from 0.22 per 1000 person-years for GFR=60–89 ml/min per 1.73 m^2 to 1.08 per 1000 person-years for GFR<30 ml/min per 1.73 m^2 (Figure 1). The age–sex-adjusted incidence rates of any cancer as well as urothelial cancer were also significantly higher at lower GFR levels (Figure 2). For breast, lung, colorectal, and prostate cancer types, there was no apparent stepwise progression observed like with renal or urothelial cancer types (Figure 2).

After adjustment for age, sex, race, socioeconomic status, comorbidities, proteinuria, hematuria, body mass index, smoking status, imaging use, health care use, and specific prescription medications, compared with GFR=60–89 ml/min per 1.73 m^2, there was an increased rate of any incident renal cancer, ranging from a 39% increased rate for GFR=45–59 ml/min per 1.73 m^2 to a more than 2-fold increased rate for GFR<30 ml/min per 1.73 m^2 (Table 2). This increased rate was of greater magnitude for clear cell renal cancer compared with nonclear cell renal cancer (Table 2). Compared with GFR=60–89 ml/min per 1.73 m^2, we also found that GFR<30 ml/min per 1.73 m^2 was associated with a 48% increased adjusted rate of urothelial cancer (Table 3). However, after adjustment for potential confounders, compared with GFR=60–89 ml/min per 1.73 m^2, GFR levels below 60 ml/min per 1.73 m^2 were not significantly associated with prostate, colorectal, lung, breast, or any cancer (Table 3). Of note, GFR=90–150 ml/min per 1.73 m^2 was associated with a reduced adjusted rate of prostate cancer and higher adjusted rates of lung and colorectal cancer compared with GFR=60–89 ml/min per 1.73 m^2 (Table 3).

DISCUSSION

Within a contemporary, community-based sample of more than 1 million adults with known kidney function, we found that lower level of GFR was independently associated with a higher risk of incident diagnosed renal cancer, even after adjustment for a broad set of potential confounders and relevant medication exposures. Subjects with GFR<30 ml/min per 1.73 m^2 had a more than 2-fold increased adjusted rate of incident renal cancer compared with those subjects who had GFR=60–89 ml/min per 1.73 m^2. We also found an association between GFR and incident urothelial cancer, although the magnitude of this association was less pronounced than observed with renal cancer. In contrast, there was no significant association between lower level of GFR and all combined cancers or prostate, breast, lung, or colorectal cancer after accounting for potential confounding factors.

Previous studies have shown an increased overall risk of cancer in persons with ESRD requiring RRT,[4–11] but the impact of less severe kidney dysfunction and cancer risk has been less certain.[12–15] In contrast to our findings, two prior studies found an association between renal function and overall cancer risk,[13,14] whereas another study found no association between CKD and cancer risk in diabetic patients.[15] When focusing on specific cancer types, three studies found that lower kidney function was associated with higher risks of renal and urothelial cancer and support our findings.[12–14] In a cohort of approximately 3600 participants, men had a significantly increased risk of cancer with lower GFR; stage 3 CKD was an independent risk factor for cancer, and this risk increased in a linear fashion with lower baseline eGFR.[14] The relation between kidney function and cancer risk was site-specific, with the most robust association seen with urinary tract (excluding prostate) and lung cancers.[14] Higher albumin-to-creatinine ratio has also been reported to be associated with a higher incidence of cancer (especially bladder and renal cancer) in adjusted analyses.[13] A separate study has also observed increased renal and urothelial cancer-specific mortality among patients with CKD.[16] Among a much larger, diverse population-based sample, we did not find GFR to be an independent predictor of overall cancer risk or specific cancers, such as prostate, breast, lung, or colorectal, after rigorously adjusting for potential confounders and excluding any incident cancers diagnosed within the first 2 years of follow-up and all GFR measurements within 3 months before a cancer diagnosis. Our findings reveal that the association of CKD and

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Table 1. Baseline characteristics stratified by level of kidney function at entry among 1,190,538 adults ages 40 years and older with no previously known cancer, dialysis, or renal transplantation

Characteristic	Overall Cohort	eGFR (ml/min per 1.73 m^2)				
		90–150	60–89	45–59	30–44	<30
N	1,190,538	309,927	704,437	131,889	34,669	9616
Age, yr						
Median age (interquartile range)	55 (48–65)	49 (44–54)	56 (49–64)	69 (60–77)	76 (68–83)	76 (67–83)
Age category, %[a]						
40–49	33.34	56.27	29.98	7.47	2.88	5.10
50–59	30.84	31.09	34.65	17.39	8.38	9.65
60–69	18.95	9.56	21.56	27.61	17.45	17.42
70–79	11.93	2.89	10.70	31.65	37.04	31.99
≥80	4.93	0.19	3.11	15.87	34.25	35.85
Sex, %						
Men	47.04	35.50	50.95	53.03	47.85	47.54
Women	52.96	64.50	49.05	46.97	52.15	52.46
Race, %						
White	58.21	45.25	60.31	71.94	75.61	71.42
Black	6.82	9.24	5.93	5.60	6.97	10.64
Native American	1.51	1.24	1.46	2.11	2.51	2.44
Asian	11.00	14.83	9.96	8.26	8.64	10.43
Other, multiple, or unknown race	22.45	29.45	22.35	12.10	6.26	5.06
Socioeconomic status, %						
Low annual household income	10.04	10.89	9.21	11.09	13.87	15.72
Low educational attainment	19.41	24.10	17.59	17.55	20.39	24.27
Medical history, %						
Heart failure	1.77	0.42	1.02	4.53	12.96	21.58
Hypertension	31.40	21.78	29.61	50.68	68.92	73.11
Diabetes	10.27	11.11	8.60	12.92	20.84	30.82
Coronary heart disease	2.44	1.04	2.09	5.14	9.07	11.57
Chronic lung disease	24.05	23.45	23.28	26.77	31.99	33.55
Chronic liver disease	0.37	0.47	0.30	0.36	0.61	0.88
Proteinuria	6.21	5.15	4.94	9.44	19.89	39.57
Hematuria						
Negative	9.33	9.58	9.91	7.19	5.15	3.48
Abnormal	6.85	7.32	6.58	6.84	7.26	10.46
Missing	83.81	83.10	83.51	85.97	87.59	86.05
Body mass index category[b]						
<25	24.48	26.00	24.33	22.79	21.92	18.87
25–29	25.06	24.34	26.04	24.10	18.37	13.87
30–39	34.04	35.03	34.60	32.10	25.37	18.22
≥40	3.30	4.01	3.20	2.62	2.07	1.72
Missing	13.12	10.63	11.83	18.39	32.28	47.32
Smoking status[b]						
Never	72.63	71.45	73.72	72.66	65.11	57.17
Current	13.78	16.91	13.28	10.03	10.59	12.10
Former	11.48	9.96	11.07	14.43	19.22	22.30
Unknown	2.11	1.68	1.92	2.88	5.08	8.43
Baseline medication use, %						
Angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker	20.92	15.34	19.04	33.35	50.82	59.58
Aldosterone receptor blocker	6.83	5.87	6.32	9.25	13.75	17.02
α-Blocker	6.30	2.55	5.70	13.29	19.82	26.98
β-Blocker	22.36	16.35	21.04	34.92	47.04	51.52
Calcium channel blocker	14.98	10.56	13.49	24.83	37.84	49.14
Statin	10.49	6.01	10.22	18.12	22.78	25.45
Nonstatin lipid-lowering drug	8.68	7.07	8.26	12.12	15.97	18.77

Table 1. Continued

Characteristic	Overall Cohort	eGFR (ml/min per 1.73 m^2)				
		90–150	60–89	45–59	30–44	<30
5-α Reductase inhibitor	0.37	0.15	0.38	0.62	0.81	0.82
Health care use during 12 months before cohort entry						
Median (interquartile range) ambulatory visits	2 (1–5)	2 (1–5)	2 (1–5)	2 (1–5)	2 (1–5)	2 (0–4)
Median (interquartile range) hospitalizations	0 (0–0)	0 (0–0)	0 (0–0)	0 (0–0)	0 (0–0)	0 (0–0)
Abdominal imaging use						
Computed tomography	2.10	2.04	1.89	2.66	3.76	5.05
Magnetic resonance imaging	0.03	0.02	0.02	0.03	0.09	0.63
Ultrasound	2.92	2.96	2.49	3.35	6.34	14.75

[a]Column percent unless otherwise specified.
[b]Closest to baseline body mass index.

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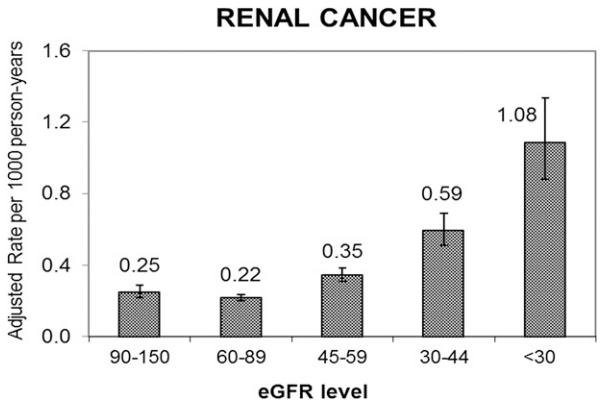

Figure 1. Age–sex-adjusted crude rate per 1000 person-years of incident renal cancer by level of eGFR among 1,190,538 adults. Subjects were 40 years and older with known renal function between 2000 and 2008.

cancer risk is site-specific for renal and urothelial cancers and does not seem to be associated with an individual's overall cancer risk.

Several possible biologic mechanisms may help to explain the association between level of kidney function and renal or urothelial cancers. Kidney dysfunction results in a state of chronic inflammation and oxidative stress,[17,18] and such an inflammatory microenvironment may play a role in cancer development.[19] Severe CKD may additionally create a relative state of immunodeficiency,[20] which could influence the development of cancer. Kidney transplant patients with medically induced immunodeficiency are known to have higher rates of cancer, and some, but not all, of that increased risk is associated with viral-induced malignancies.[11] Certain medical therapies[21–26] used in patients with CKD have been suggested to increase cancer risk, but adjustment for differential longitudinal use of these medications did not materially affect the association between level of GFR and renal or urothelial cancer. These mechanisms and others deserve additional study to better define the link between kidney function and site-specific cancer risk. The stronger association of GFR and clear cell type renal cancer compared with nonclear cell renal cancer may further elucidate the biologic underpinnings of the association of CKD and renal cancer.

Our study has several notable strengths. We evaluated approximately 1.2 million adults with known renal function and no prior known cancer, representing the largest study to date exploring the relation between CKD and cancer risk. Our study focused on patients across a broad spectrum of kidney function and excluded patients with known ESRD. We ascertained longitudinal measures of GFR based on the recommended Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation[3,27] and statistically adjusted for a large set of potential time-updated confounders as well as a comprehensive list of medications that potentially impact cancer risk. Another key strength of this study was our ability to minimize the impact of detection bias by additional adjustment for hematuria and receipt of abdominal imaging. None of the previous studies have been able to adjust for these important confounding factors. Importantly, in a subgroup analysis not shown here, we found an association between eGFR and renal cancer among those subjects with nonlocalized cancer who did not have abdominal imaging; arguably, this group is the least susceptible to detection bias, adding to the evidence supporting an etiologic role of reduced kidney function in renal carcinogenesis. Additionally, we comprehensively ascertained the occurrence of cancer through use of a regional cancer registry linked to Surveillance, Epidemiology, and End Results (SEER).[28]

This study had certain limitations. We cannot completely exclude the impact of residual confounding. For example, information on family history of cancer, severity of comorbid conditions, and drug dosage was unavailable. However, we controlled for the presence or absence of major confounders and longitudinal exposure to the most common relevant medications that may impact cancer risk, and the strong increased risk of renal cancer (and to a lesser degree, urothelial cancer) argues against missing a major confounder. Data were also unavailable for selected over-the-counter analgesics and toxin exposure. Phenacetin, an analgesic that has been off the market for more than three decades, has been closely linked to both urothelial and renal cancer,[29,30] but it is very unlikely to have been used by our cohort members in our contemporary study period. However, we recognize that recent data raise some concern about the association of renal cancer and nonaspirin, nonsteroidal anti-inflammatory drug use.[31] Any of the serum creatinine-based GFR estimating

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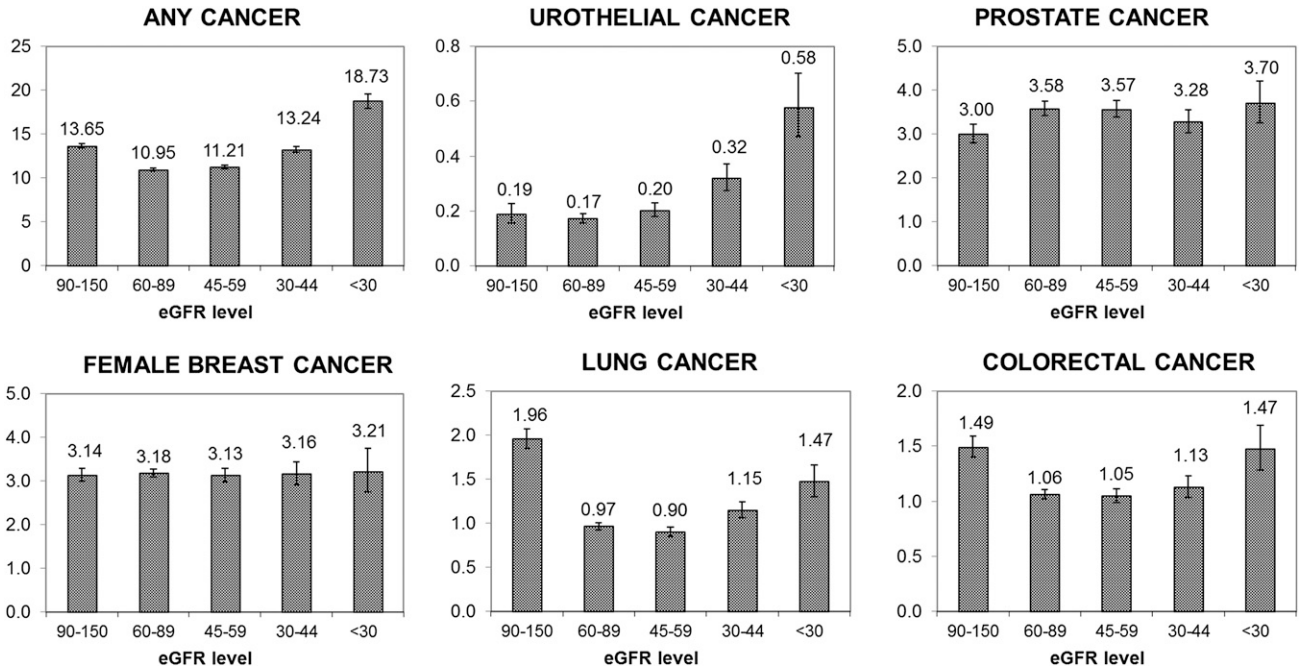

Figure 2. Age–sex-adjusted crude rate per 1000 person-years of any incident cancer and specific cancers (urothelial, prostate, breast, lung, and colorectal) by level of eGFR among 1,190,538 adults. Subjects were 40 years and older with known renal function between 2000 and 2008.

Table 2. Multivariable association between measures of kidney function and risk of incident renal cancer (overall and by type)

eGFR (ml/min per 1.73 m^2)	Cancer Cases Number and Adjusted HR (95% CI)[a]					
	Any Renal Cell Carcinoma[b]		Clear Cell Renal Carcinoma[c]		Nonclear Cell Renal Carcinoma[c]	
	N	HR (95% CI)	*N*	HR (95% CI)	*N*	HR (95% CI)
90–150	241	1.08 (0.93 to 1.26)	116	1.12 (0.9 to 1.39)	40	1.37 (0.93 to 2.00)
60–89	826	Reference	383	Reference	117	Reference
45–59	395	1.39 (1.22 to 1.58)	165	1.29 (1.06 to 1.58)	52	1.24 (0.87 to 1.76)
30–44	196	1.81 (1.51 to 2.17)	80	1.71 (1.29 to 2.26)	25	1.55 (0.96 to 2.49)
<30	92	2.28 (1.78 to 2.92)	28	1.68 (1.09 to 2.58)	6	0.92 (0.38 to 2.21)

HR, hazard ratio; 95% CI, 95% confidence interval.
[a]Final models adjusted for age, sex, race, income, education, smoking, body mass index, prior health care use, proteinuria, hematuria, imaging use, specific comorbidities, and prescription drug use as listed in Table 1.
[b]List of histology codes: 80003, 80103, 80503, 81403, 82113, 82553, 82603, 83103, 83123, 83163, 83173, 83183, 83193, 83203, and 83233.
[c]Histology codes: clear cell carcinomas, 83103; nonclear cell carcinomas, 80503, 82603, 83163, 83173, 83183, 83193, and 83203.

equations may provide imperfect measurements of GFR, introducing the potential for misclassification of kidney function level. We sought to minimize this misclassification by using longitudinal, time-updated measurements of GFR calculated by the CKD-EPI equation[3,27] and isotope dilution mass spectrometry–traceable serum creatinine measures, with a median of four outpatient, nonemergency department serum creatinine measurements per subject. Given that patients with worse kidney function may be more closely followed clinically and possibly, more likely to have incident cancer detected, we adjusted for prior medical resource use, which did not materially affect the relations between level of kidney function and incident renal or urothelial cancer. We also limited potential effect cause by excluding any incident cancers diagnosed during the first 2 years of follow-up as well as any GFR measures performed within 3 months before an incident cancer diagnosis. Finally, although our study sample was large and diverse, results may not be fully generalizable to other populations and practice settings.

In conclusion, we found that lower GFR level was an independent risk factor for incident renal and urothelial cancer but not prostate, colorectal, lung, breast, or all cancers combined. If GFR is associated with an increased risk of renal and urothelial cancer, then it could have implications for directing cancer screening efforts in select populations. Currently, there are no evidence-based cancer screening recommendations tailored for patients with CKD. Additional studies are needed to elucidate the reasons for this association and delineate the potential use of targeted cancer screening in patients with CKD.

Table 3. Multivariable association between measures of kidney function and risk of incident cancer (overall and by cancer type)

eGFR (ml/min per 1.73 m^2)	Cancer Cases Number and Adjusted HR (95% CI)[a]											
	Any Cancer		Urothelial[b]		Prostate		Breast		Lung		Colorectal	
	N	HR (95% CI)	N	HR (95% CI)	N	HR (95% CI)	N	HR (95% CI)	N	HR (95% CI)	N	HR (95% CI)
90–150	10,271	1.04 (1.01 to 1.06)	114	0.95 (0.77 to 1.17)	1014	0.76 (0.71 to 0.81)	2100	0.98 (0.92 to 1.03)	1414	1.32 (1.22 to 1.43)	1066	1.18 (1.09 to 1.28)
60–89	39,105	Reference	727	Reference	9001	Reference	5966	Reference	4258	Reference	3728	Reference
45–59	15,151	1.00 (0.98 to 1.02)	439	1.12 (0.99 to 1.27)	3422	1.02 (0.98 to 1.06)	1733	0.96 (0.91 to 1.02)	1849	0.96 (0.90 to 1.02)	1578	0.98 (0.91 to 1.04)
30–44	6116	1.05 (1.01 to 1.08)	267	1.42 (1.21 to 1.67)	887	0.98 (0.91 to 1.06)	617	0.92 (0.84 to 1.01)	865	1.01 (0.92 to 1.1)	653	1.02 (0.92 to 1.12)
<30	2218	1.05 (0.99 to 1.11)	126	1.48 (1.19 to 1.85)	259	1.02 (0.89 to 1.18)	161	0.85 (0.71 to 1.01)	285	0.80 (0.68 to 0.93)	223	1.05 (0.89 to 1.24)

HR, hazard ratio; 95% CI, 95% confidence interval.

[a]Final models adjusted for age, sex, race, income, education, smoking, body mass index, prior health care use, proteinuria, specific comorbidities, and prescription drug use as listed in Table 1. Urothelial cancer final model is additionally adjusted for hematuria.

[b]Urothelial cancers included transitional cell carcinoma (histology codes 81203 and 81303).

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CONCISE METHODS

Study Sample

We identified all persons with measured serum creatinine concentration who are receiving care within Kaiser Permanente Northern California, a large integrated health care delivery system providing comprehensive care to >3.2 million persons in the San Francisco and greater Bay area. The Kaiser Permanente Northern California population is highly representative of the surrounding local and statewide population, except for slightly lower representation at the extremes of age and incomes.[32] The analytic sample included members who were ages 40 years and older, had one or more outpatient, nonemergency department serum creatinine measurements between January 2000 and December 2008, and had ≥12 months of continuous health plan membership before cohort entry. Patients were excluded if they had unknown sex or preexisting cancer diagnosis (except for localized, nonmelanoma skin cancer) identified in the Kaiser Permanente Cancer Registry,[28] which also links to the SEER cancer registry.[33] We also excluded patients with a documented history of hemodialysis, peritoneal dialysis, or renal transplantation before entry based on a comprehensive health plan ESRD treatment registry.[34]

The Institutional Review Board of the Kaiser Foundation Research Institute approved this study. Waiver of informed consent was obtained because of the nature of the research.

Measurement of Renal Function

The primary predictor of interest was level of kidney function as determined by eGFR at baseline and during follow-up. GFR was calculated using all available outpatient, nonemergency department serum creatinine concentration values derived from isotope dilution mass spectrometry–traceable assays using the CKD-EPI equation and incorporated as time-varying values in our analyses. The CKD-EPI equation has been shown to be more accurate than the Modification of Diet in Renal Disease study estimating equation, especially when estimating higher levels of GFR.[3,27] GFR values (in units of milliliters per minute per 1.73 m^2) were categorized according to the modified National Kidney Foundation classification system for CKD: ≥90, 60–89, 45–59, 30–44, 15–29 and <15 ml/min per 1.73 m^2 not requiring dialysis. Based on available sample size, we combined the latter two groups into a single category (<30 ml/min per 1.73 m^2).

Follow-Up and Outcomes

Follow-up occurred through December 31, 2008, which was the latest date in which complete vital status data were available at the time of analysis. Patients were censored at the end of follow-up, initiation of dialysis, date of renal transplantation, disenrollment from the health plan, or death. Death was identified from health plan administrative data (including proxy reporting), California death certificate data, and Social Security Administrative vital status files.[35,36]

The primary outcome of interest was occurrence of incident renal cancer during the study period identified from the Kaiser Permanente Cancer Registry. Renal cancer histology codes used in our dataset were in compliance with the World Health Organization International Classification of Diseases for Oncology international standard for primary site and histology (December 2009; www.seer.cancer.gov/

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icd-o-3/index.html). Specific histology codes used are listed in Table 2. In secondary analyses, we evaluated the association of renal function with any incident cancer diagnosis (excluding nonmelanoma skin cancers) and other specific cancer types, including urothelial, prostate, breast, lung, and colorectal. Specific urothelial histology codes used are listed in Table 3.

Covariates

Information on age, sex, and race was obtained from health plan databases. Socioeconomic characteristics were assigned using US Census-block data.[32,34] Subjects living in a Census-block area where median annual household income was less than $35,000 were classified as low income. Low educational attainment was also categorized according to US Census-block data and defined as those individuals living in an area where more than 25% of residents over age 25 years have less than a 12th grade education level.

The presence of proteinuria and hematuria was based on the urine dipstick results found in health plan laboratory databases. As previously described,[34] proteinuria was defined as a urine dipstick protein result of 1+ or greater (approximately 30 mg/dl or greater) in the absence of possible urinary tract infection (*i.e.*, concomitant positive urine nitrite or esterase).

We identified baseline and time-updated comorbidities within our cohort by methods previously described using health plan databases for inpatient and ambulatory diagnoses, laboratory results, and prescription medications.[34,37–40] Coexisting illnesses of interest included diabetes mellitus, hypertension, coronary heart disease, heart failure, and chronic lung or liver disease. Body mass index category[41] and tobacco history were ascertained from ambulatory visit databases. Measures of health care use before cohort entry included the number of ambulatory visits and hospitalizations during the 12 months before index date. We identified the total number of radiology studies (computed tomography, magnetic resonance imaging, and ultrasound of the abdomen) obtained during follow-up. Longitudinal medication use was ascertained from ambulatory pharmacy records for common medications potentially associated with cancer risk using previously described algorithms.[37,42] These medications included oral antihypertensive agents[21,22] (β-blockers, angiotensin-converting enzyme inhibitors, angiotensin II receptor blockers, α-blockers, calcium channel blockers, and aldosterone receptor antagonists), nonstatin lipid-lowering agents, statins,[24,25] and 5-α reductase inhibitors.[23,26]

Analytic Approach

Analyses were performed using SAS software, version 9.2 (SAS Institute, Cary, NC). Baseline characteristics were compared across GFR categories using ANOVAs for continuous variables and chi-squared tests for categorical variables. We calculated rates (per 1000 person-years) of incident cancer by GFR category, overall, and specific cancer type. We conducted extended Cox regression models to examine the independent association between measures of kidney function and risk of any incident cancer and specific cancer type after adjustment for potential confounders. Final models were adjusted for age, sex, race, socioeconomic status, smoking status, body mass index, health care use, proteinuria, comorbid conditions (heart failure, hypertension, diabetes mellitus, coronary heart disease, chronic lung disease, and chronic liver disease), and longitudinal exposure to targeted prescription drugs. The final urothelial cancer model was additionally adjusted for hematuria, whereas the final model for renal cell cancer was also adjusted for hematuria and receipt of abdominal imaging (Table 1). To further evaluate any potential association of kidney function and renal cell cancer, we fit the above-described model for conventional clear cell renal cancer and nonclear cell renal cancer. We separately examined associations using baseline and time-updated measures of kidney function. For all models, we used a sandwich estimate of the variance–covariance matrix to obtain SEMs accommodating the clustering of observations on subjects.[43] Because of concern for potential effect cause (or reverse causation) for the association between GFR and risk of cancer, we excluded any incident cancers that occurred during the first 2 years of follow-up and any GFR measures performed within 3 months before an incident cancer diagnosis during follow-up.

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DISCLOSURES

None.

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See related editorial, "CKD and Risk of Renal Cell Carcinoma: A Causal Association?," on pages 2147–2148.